



Week 1

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Mechanics

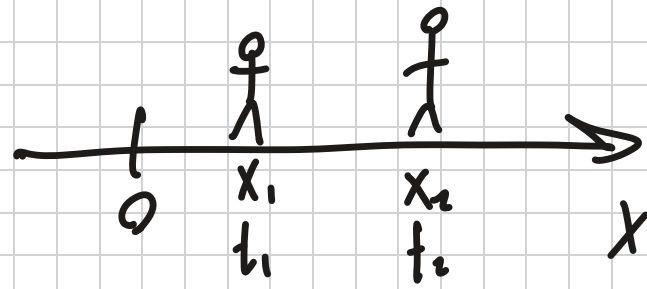
Kinematics

description of motion

Dynamics

tries to explain motion
(Newton's Law)

2. Kinematics of motion in 1-D



- x axis with origin point
- 1 space dimension x
- 1 time dimension t

Units of measurement

distance $\rightarrow m$

time $\rightarrow s$

Displacement: change in position

$$\Delta X = X_2 - X_1$$

Can be positive or negative

distance traveled $\neq$ displacement
positive quantity

Average velocity

$$V_{avg} = \frac{\Delta X}{\Delta t} = \frac{X_2 - X_1}{t_2 - t_1}$$

$$\langle X \rangle = m$$

$$\langle t \rangle = s$$

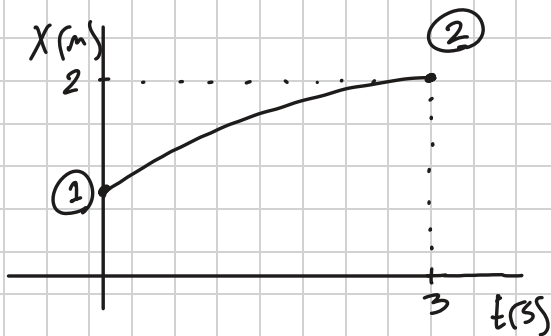
$$\langle V \rangle = \frac{\langle X \rangle}{\langle t \rangle} = \frac{m}{s}$$

velocity

distinction between velocity (can be -)
and speed (always positive)

$$\text{Speed} = \frac{\text{distance traveled}}{\text{time elapsed}} > 0$$

example 1



$$X_1 = 1m, t_1 = 0s$$

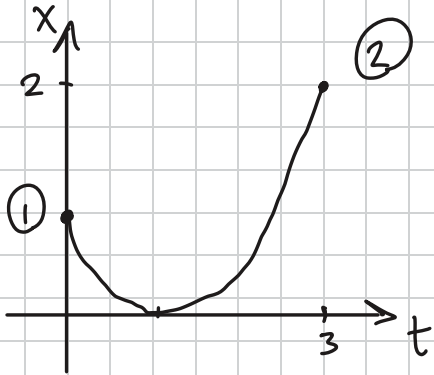
$$X_2 = 2m, t_2 = 3s$$

$$\Delta X = X_2 - X_1 = 1m$$

$$\Delta t = t_2 - t_1 = 3s$$

$$V_{ave} = \frac{\Delta X}{\Delta t} = \frac{1m}{3s} = 0.3m/s$$

$$\begin{aligned} \text{Speed}_{ave} &= \frac{\text{distance traveled}}{\text{time}} \\ &= 0.3m/s \end{aligned}$$



Value same as before

distance traveled = 3 m

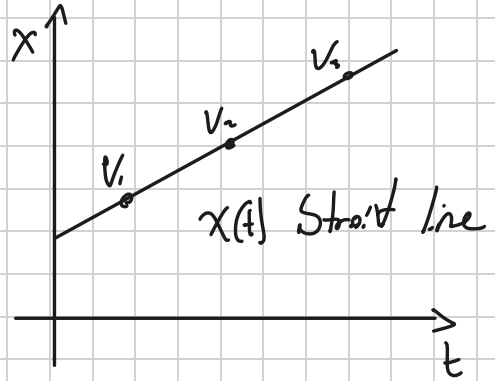
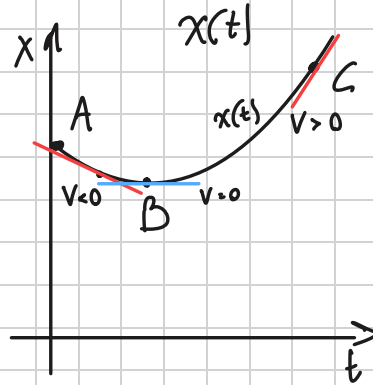
$$\text{Speed}_{\text{ave}} = \frac{3 \text{ m}}{3 \text{ s}} = 1 \text{ m/s}$$

2.2 instantaneous velocity

$$V_{\text{ave}} = \frac{\Delta x}{\Delta t}$$

$$V = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} \quad \text{instantaneous velocity}$$

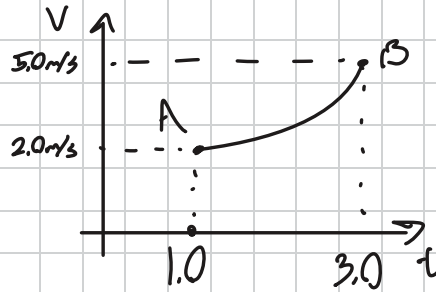
$$V = \frac{dx}{dt} \quad \text{derivative of the pos with respect to time}$$



$$V_1 = V_2 = V_3 = V \\ \text{constant}$$

2.3 acceleration

$$a_{\text{ave}} = \frac{V_2 - V_1}{t_2 - t_1} = \frac{\Delta V}{\Delta t}$$



$$\langle a \rangle = \frac{\langle v \rangle}{\langle t \rangle} = \frac{\text{m/s}}{\text{s}} = \text{m/s}^2$$

$$\begin{aligned} a_{\text{ave}} &= \frac{5.0 \text{ m/s} - 2.0 \text{ m/s}}{3.0 \text{ s} - 1.0 \text{ s}} \\ &= \frac{3.0 \text{ m/s}}{2.0 \text{ s}} \\ &= 1.5 \text{ m/s}^2 \end{aligned}$$

Week 2